MAT 4233 — Weekly Schedule

Modern Abstract Algebra · Fall 2026 · UT San Antonio

Texts

Primary text

- **LiZh** — Li & Zhao, *Introduction to Abstract Algebra*, Les Ulis: EDP Sciences, 2022. ISBN 978-2-7598-2915-6. DOI: 10.1051/978-2-7598-2916-3

Supplementary reading

- **Childs** — Childs, *A Concrete Introduction to Higher Algebra*, 3rd ed., Springer, 2009. ISBN 978-0-387-74527-5. DOI: 10.1007/978-0-387-74725-5
- **Rosebrock** — Rosebrock, *Visual Group Theory*, Springer, 2024. ISBN 978-3-662-69364-3. DOI: 10.1007/978-3-662-69365-0
- **Beardon** — Beardon, *Algebra and Geometry*, Cambridge University Press, 2005. ISBN 978-0-521-81362-4. DOI: 10.1017/CBO9780511800436

Handouts & papers

- **Conrad** — Conrad, *The Gaussian Integers*

Schedule

Date	Topic	Reading
Thu 20 Aug	Lecture — Course launch. The integers: division algorithm, gcd, Bezout	Childs Ch. 3
Tue 25 Aug	Lecture — Congruence and the ring Zn	Childs Ch. 5 & 6
Thu 27 Aug	Lecture — Binary operations; Cayley tables; isomorphic binary structures	LiZh §1.1-1.2
Tue 1 Sep	Lecture — Plane isometries and finite symmetry groups: cyclic and dihedral	Rosebrock Ch. 1; Beardon §3.1-3.4; E-handout: Complex geometry (passphrase in Canvas)
Thu 3 Sep	Lecture — Groups: axioms and first properties	LiZh §1.3
Tue 8 Sep	**Q&A** (**Due**: Set 1)	Problem Set 1 (PDF); Activity: Symmetry Groups and Cayley Graphs (Colab, ungraded)
Thu 10 Sep	Lecture — Subgroups; examples	LiZh §1.4
Tue 15 Sep	Lecture — Cyclic groups; the order of an element; generators	LiZh §1.5
Thu 17 Sep	Lecture — Generating sets	LiZh §1.6
Tue 22 Sep	Lecture — Permutation groups; cycle notation; Cayley's theorem	LiZh §2.1
Thu 24 Sep	Lecture — Alternating groups; the sign of a permutation; A4 and the tetrahedron	LiZh §2.2
Tue 29 Sep	**Q&A** (**Due**: Set 2)	Problem Set 2 (PDF); Activity: Permutation Puzzles (Colab, ungraded)
Thu 1 Oct	**Exam** — Midterm Exam 1 — Sets 1 and 2	
Tue 6 Oct	Lecture — Cosets; Lagrange's theorem; index	LiZh §3.1
Thu 8 Oct	Lecture — Homomorphisms: kernel, image, and first properties	LiZh §3.3

Date	Topic	Reading
Thu 15 Oct	Lecture — Normal subgroups; quotient groups	LiZh §3.4
Tue 20 Oct	Lecture — Normal subgroups and quotient groups, continued	LiZh §3.4
Thu 22 Oct	Lecture — The fundamental homomorphism theorem; groups of small order	LiZh §3.4
Tue 27 Oct	**Q&A** (**Due**: Set 3)	Problem Set 3 (PDF); Activity: Cosets, Normality and Quotient Groups (Colab, ungraded)
Thu 29 Oct	Lecture — Rings and subrings; units and zero divisors; integral domains and fields	LiZh §4.1-4.2; LiZh §4.6 (Euler's and Fermat's theorems, already met via Lagrange in Set 3)
Tue 3 Nov	Lecture — Ring homomorphisms, ideals and quotient rings; fundamental homomorphism theorem	LiZh §4.5
Thu 5 Nov	Lecture — Ideal theory: principal, prime and maximal ideals	LiZh §4.7
Tue 10 Nov	Lecture — Irreducibles and primes; unique factorization domains; principal ideal domains	LiZh §5.1-5.2
Thu 12 Nov	Lecture — Euclidean domains; the Euclidean algorithm in Z and in k[x]	LiZh §5.3
Tue 17 Nov	Lecture — Multiplicative norms; the Gaussian integers Z[i]; Fermat's Two Squares theorem	LiZh §5.5; Conrad (statements and examples; proofs optional)
Thu 19 Nov	**Q&A** (**Due**: Set 4)	
Tue 24 Nov	Lecture — Sums of three and four squares; Z[sqrt(-5)] and the failure of unique factorization	Instructor notes
Tue 1 Dec	*Review* — Review — the exam pool	
Thu 3 Dec	**Exam** — Midterm Exam 2 — Sets 3 and 4	